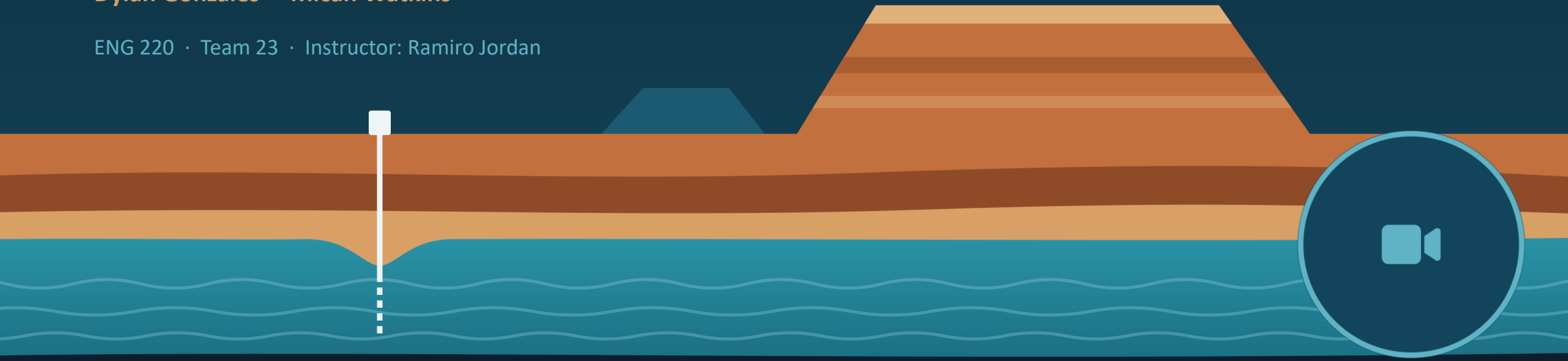
Beneath the Mesa

Assessing Aquifer and Well Water Resources Across Sandoval County, New Mexico

Petar Djordjevic (PM) · Meckrem Yousef · James Wauneka
Dylan Gonzales · Micah Watkins

ENG 220 · Team 23 · Instructor: Ramiro Jordan



The project at a glance

What we're building

- A screening study of water resources across Sandoval County, with Cuba as the focus
- An interactive app, a written report, and an open GitHub repository

What it does

- Brings together public well, aquifer, and river data
- Shows what's known about Cuba's groundwater, and where the data runs out

Who benefits

- Cuba residents and village leaders
- Rural communities that rely on wells
- Agencies deciding where to monitor next

Peace engineering
Clear, shared information about a scarce resource helps communities make fair water decisions together.



Background

- Where Sandoval County's water comes from: wells, rivers, and acequias
- Why the county spans several aquifer systems
- The concern: declining groundwater near Cuba
- What this project set out to check



Figure caption and source

Research questions

MAIN QUESTION

Is Cuba's aquifer running out, and do public records say enough to tell?

1. **Groundwater levels**
   How have groundwater levels changed at monitored wells in Sandoval County?

2. **Cuba's aquifers**
   How do the Ojo Alamo Sandstone and San Jose Formation compare near Cuba?

3. **Coverage and data gaps**
   Where is monitoring coverage strong, and where are the biggest data gaps?

Deliverables

Deliverable	What it is	Owner	Due
Streamlit app	Interactive maps and charts of the data	Meckrem	**Nov 14**
Final report	Methods, findings, and figures in writing	Petar	**Nov 14**
Presentation slides	This deck, presented in class	Petar	**Nov 14**
Individual videos	Each member presents their own part	Everyone	**Nov 15**
Documentation site	How each stage was done, with links to everything	Petar	**Nov 16**
GitHub repository	Raw and cleaned data, scripts, and app code	Everyone	**Nov 16**

FINAL DUE

Mon, Nov 16

Nov 14
App, report, and slides

Nov 15
Individual videos



Data

Groundwater levels

USGS: 9 wells, 1998–2026. NM Bureau of Geology: 87 wells, 1997–2019.

Aquifer tests

NM State Engineer: 179 tests from 1930 to 2008

Rio Puerco at Cuba

USGS gage 08332525: 10 streamflow visits from 2019 to 2026

Well logs and geology

NM State Engineer: 6 well logs. USGS and NMGS maps of the two formations.

Full source links are on the last slide and in the project documentation.



Methods

Collect

Dylan Gonzales

One or two lines on what this step did

Clean

James Wauneka

One or two lines on what this step did

Analyze

Micah Watkins

One or two lines on what this step did

Present

Meckrem Yousef

One or two lines on what this step did

Report

Petar Djordjevic

One or two lines on what this step did

Tools Python · pandas · Plotly · Streamlit · GitHub

Data collection

Dylan Gonzales

Data acquisition and dataset justification

What I did
The part of the project you were responsible for

How I did it
Tools, sources, and main steps

What I learned
A challenge you hit and how you handled it

Screenshot of your work

A data source page or the raw dataset

Figure caption and source

Data cleaning

James Wauneka

Data cleaning and preprocessing

What I did
The part of the project you were responsible for

How I did it
Tools, sources, and main steps

What I learned
A challenge you hit and how you handled it

Screenshot of your work

The cleaning script, or the data before and after

Figure caption and source

Analysis and visualizations

Micah Watkins

Analysis code and visualizations

What I did
The part of the project you were responsible for

How I did it
Tools, sources, and main steps

What I learned
A challenge you hit and how you handled it

Screenshot of your work

A chart from the analysis

Figure caption and source

Streamlit app

Meckrem Yousef

GitHub workflow and the Streamlit app

What I did
The part of the project you were responsible for

How I did it
Tools, sources, and main steps

What I learned
A challenge you hit and how you handled it

Screenshot of your work

A page of the live app

Figure caption and source

Report and documentation

Petar Djordjevic

Project manager: report, slides, and video

What I did
The part of the project you were responsible for

How I did it
Tools, sources, and main steps

What I learned
A challenge you hit and how you handled it

Screenshot of your work

The documentation site or a page of the report

Figure caption and source

Results

Chart, map, or app screenshot

Copy this slide for each result

Figure caption and source

— ft

Key number and what it measures

What this shows

Two or three sentences on the main pattern in the chart.

Findings

What the data show

- Measured results, with numbers

What they suggest

- Likely explanations, stated as possibilities

Too limited to tell

- Questions the data can't answer yet

Bottom line
One sentence that answers the main question



Limitations

- Areas with few or no monitoring wells
- Short or irregular water-level records
- Data the project couldn't get
- What a stronger study would need

Data coverage map

Where monitoring is dense, sparse, or absent

Figure caption and source

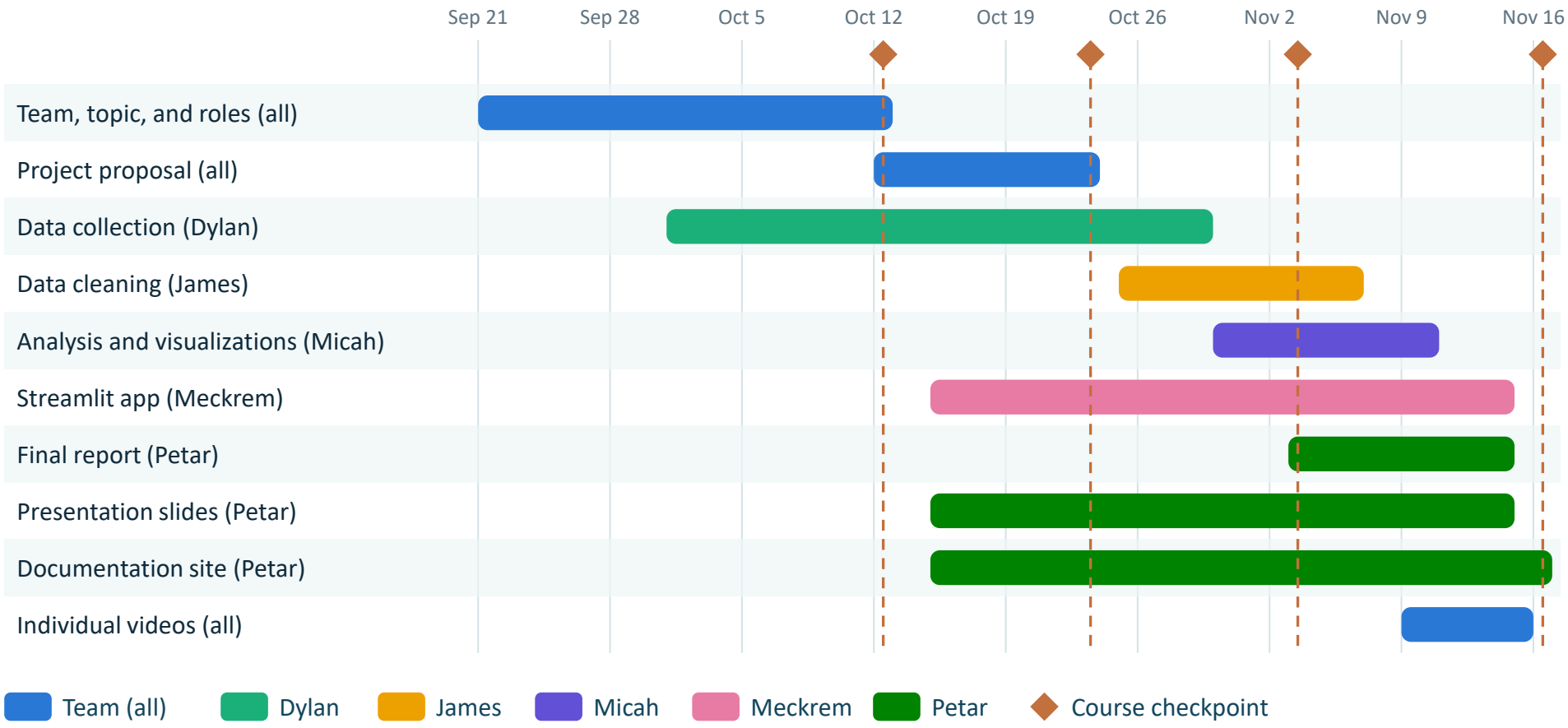
Risks and how we reduce them

Risk	How we reduce it
No records from Cuba's own wells	Ask the Village of Cuba and NM OSE for records; label every result as Cuba, county, or regional
Sparse, uneven monitoring	Map where coverage is thin, trend only wells with long records, and state the gaps
Errors in the raw data	Shared cleaning rules, a rerunnable script, and checks against the raw files
Overstating conclusions	Keep what the data show, suggest, and can't tell separate in every finding
Tight final weeks	Write the report and slides alongside the app, and track every deadline on the timeline

Guardrails from our proposal

- Treat the concern about Cuba's aquifer as a question, not a conclusion
- Keep Cuba as the main case study to stay on schedule
- Use public data anyone can recheck

Timeline and milestones



Milestones

- **Mon, Oct 12**
  First quarter · done
- **Fri, Oct 23**
  Midpoint · done
- **Tue, Nov 3**
  Three-quarter
- **Mon, Nov 16**
  Final due

Explore the data, code, and app

GitHub repository
github.com/pdjordj11/eng_220_aquifer_project

Streamlit app
eng220aquiferproject.streamlit.app

Documentation
pdjordj11.github.io/eng_220_aquifer_project

Datasets
github.com/pdjordj11/eng_220_aquifer_project/tree/main/data

